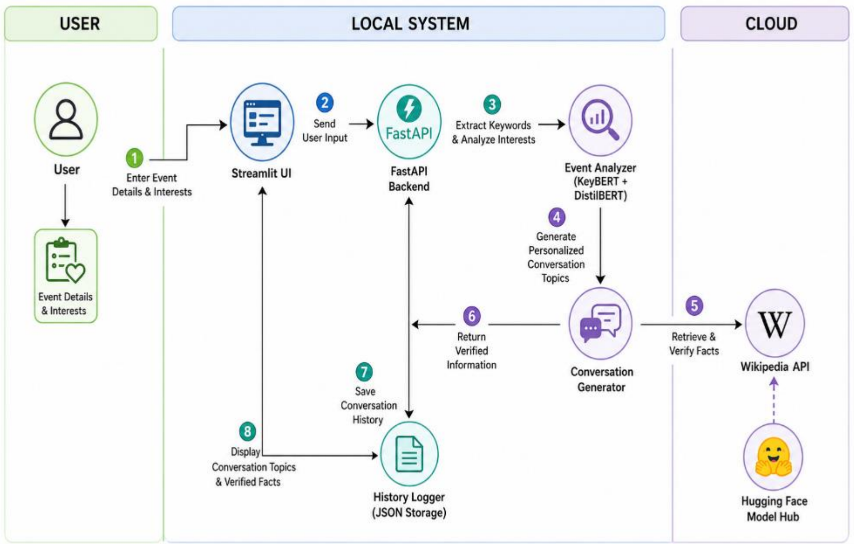
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	05 July 2026
Team ID	
Project Name	Personalised Networking Assistant
Maximum Marks	4 Marks

Technical Architecture:

The **Personalized Networking Assistant** follows a client-server architecture built using **Streamlit** for the frontend and **FastAPI** for the backend. Users enter event details and personal interests through the Streamlit interface. The backend processes the input using **KeyBERT with DistilBERT** to extract keywords and generate personalized conversation topics. The generated content is then verified using the **Wikipedia API** for fact-checking. Finally, the application displays the conversation suggestions, verified information, and stores conversation history and user feedback in a **JSON** file for future reference.

Example: Order processing during pandemics for offline mode



Guidelines

The **Personalized Networking Assistant** uses a client-server architecture. The **local infrastructure** includes the Streamlit frontend, FastAPI backend, KeyBERT with DistilBERT, and JSON data storage. The **cloud infrastructure** includes the Hugging Face Model Hub and the Wikipedia API for fact-checking.

User input is processed to generate personalized conversation topics, verified using the Wikipedia API, and the results are stored and displayed to the user.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interface for users to enter event details and view results	Streamlit
2.	Application Logic-1	Processes user requests and manages application workflow	FastAPI (Python)
3.	Application Logic-2	Analyzes event details and extracts keywords	KeyBERT + DistilBERT
4.	Application Logic-3	Generates personalized conversation topics and performs fact-checking	Python
5.	Database	Stores conversation history and user feedback	JSON File
6.	Cloud Service	Provides pre-trained language model	Hugging Face Model Hub
7.	File Storage	Stores application data locally	Local File System
8.	External API-1	Verifies generated information	Wikipedia API
9.	External API-2	Not Used	—
10.	Machine Learning Model	Keyword extraction for conversation generation	DistilBERT (via KeyBERT)
11.	Infrastructure (Server / Cloud)	Local deployment with external cloud services	Local System + Cloud APIs

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Open-source frameworks used for application development	Streamlit, FastAPI, KeyBERT, DistilBERT, Python
2.	Security Implementations	Input validation and secure communication between application components	FastAPI Validation, HTTPS

S.No	Characteristics	Description	Technology
3.	Scalable Architecture	Modular client-server architecture supporting future enhancements	Streamlit, FastAPI
4.	Availability	Application uses reliable cloud services for AI model and fact-checking	Hugging Face Model Hub, Wikipedia API
5.	Performance	Fast request processing with lightweight backend and efficient AI models	FastAPI, KeyBERT, DistilBERT